



GamemasterScrawl (5)

Parker Hoffman

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GamemasterScrawl: An Overview

GamemasterScrawl is a web browser-based virtual tabletop mapping tool designed to give Gamemasters and their players a shared, accurate, live view of the battlefield in a 3-D Space

- **Opportunity:** Table-Top Role-Playing Games (TTRPGs) typically use a map system. However, most TTRPG systems allow for movement in all 3 dimensions. Sadly, most TTRPG map software only supports 2 dimensions. This can lead to confusion and disagreements at the table, as use of the third dimension can easily be overlooked in the heat of the moment in combat.
- **Solution:** GamemasterScrawl will be a TTRPG map system that is built with a 3D approach. It will be a browser-based application that supports live updates to multiple users simultaneously. It will also be hosted on a single client's device, which reduces server overhead on the business side. This will feature 3D space for all the maps, both in creation and interaction. This allows for players and the Gamemaster to view the battlefield from many more angles than a 2D map could provide, increasing the possibility for the creative ideas that TTRPGs are best known for. In addition, a user management system will be included, as well as an entity (called token) management system.
- **Competitive Advantage:** Currently, there is only one other competitor in the 3D TTRPG Map system space, Sigil VTT from Wizards of the Coast, and that competitor is scheduled to be sunset in October of 2026. As such, there is a gap opening up in the market that GamemasterScrawl can fill. In addition, the only other competitor is specific to a TTRPG system, Dungeons and Dragons, whereas GamemasterScrawl is designed to be TTRPG system-agnostic, which allows it to appeal to a much more broad demographic compared to Sigil VTT.

GamemasterScrawl aims to fill this emerging gap in the market by delivering a simple, out-of-the-box solution for a TTRPG group to users to manage maps in a 3-Dimensional Space

Project Definition Document (PDD)

Why am I doing this?	Purpose
	Table-Top Role-playing Games (TTRPGs) allow for a group of friends to meet up and have a fun time imagining and role-playing various characters and interactions in a large variety of scenarios. One way to keep order in this controlled chaos is to have a well-maintained and accurate central map for interacting with the fictional world. Several solutions for managing this map are high-quality, but are designed with a purely 2-dimensional vision. However, many TTRPG systems allow for movement in a 3-D space. Representing 3-D movement in these solutions is difficult, due to the 2-D top-down view of the map these software solutions provide. As such, it can lead to infighting and make these scenarios much more clunky and undesirable. Thus, the purpose of this project is to fill this niche: Making a TTRPG-system agnostic program that focuses on movement in a 3-D space. This will allow both the players and the Gamemaster to quickly and efficiently deal with these scenarios and adjust the game accordingly, without the need for undoing events or other mishaps due to a lack of the third dimension.
What goal(s) does this project support, what problems are solved, what needs are met?	Goals & Objectives
• Develop a Browser-based 3-D virtual tabletop system • Support any TTRPG system, not just Dungeons and Dragons • Provide intuitive ways to interact with and edit the 3-D space • Enhance player/Gamemaster immersion and spatial awareness	
How does this project fit with other solutions that available?	Scope, Project Context, Project Dependencies
This product will be a standalone solution to this problem. I plan on no integrations with similar products or other products in the space, with the exception of importing of some assets (Images/textures) for use in the 3-D space. This product will: • Use browser-based Clients 3-D environment, And item manipulation in this environment • Allow map Creation • Utilize modern Browser Technology This product will not: • Implement AI Features on launch	

What is the expected benefit from this project?	Expected Benefits, Business Case, Value, Success Criteria
- Streamlined TTRPG Gameplay - Enhanced Player and Gamemaster immersion - All parties' spatial awareness has been improved	
What am I going to do?	Scope
- Develop/Utilize a lightweight 3-D rendering system for browsers - Allow basic user management for Gamemasters - Implement Real-Time synchronization between all clients via Sockets - Create a system for 3-D map Creation - Use the above system for 3-D space manipulation	
Who is affected by this and who must be involved?	Stakeholders
- Gamemasters - Players - The Computer Science Department	
How will we know when I am done and whether the project was successful?	Success Criteria
- The program supports 3-D manipulation of a space - Users can create custom maps - Updates to all clients in a reasonable speed (>200ms) - Works on all major browsers (Chromium, Gecko, Safari) - Minimal crashing during use - User Management/Permissions are fully functional - Data is correctly saved between uses - Data is managed securely	
Completed by:	Parker Hoffman
Date:	1/24/2026
Project Title / ID Number	GamemasterScrawl/5
Approved by:	Dr. Cheri A. Coder

GamemasterScrawl: Technical Requirements Document (3) (Parker Hoffman)

Project Name:	GamemasterScrawl(5)
Date:	4/20/2026
Prepared by:	Parker Hoffman
Document status (draft, proposed, validated or approved)	Approved

1. Introduction

This document contains the system requirements for GamemasterScrawl. These requirements have been derived from several sources, including *Foundry Virtual Tabletop*, Mozilla, Three.js

1.1 Purpose of This Document

This document is intended to guide development of GamemasterScrawl. It will go through several stages during the course of the project:

1. **Draft:** The first version, or draft version, is compiled after requirements have been discovered, recorded, classified, and prioritized.
2. **Proposed:** The draft document is then proposed as a potential requirements specification for the project. The proposed document should be reviewed by several parties, who may comment on any requirements and any priorities, either to agree, to disagree, or to identify missing requirements. Readers include end-users, developers, project managers, and any other stakeholders. The document may be amended and re-proposed several times before moving to the next stage.
3. **Validated:** Once the various stakeholders have agreed to the requirements in the document, it is considered validated.
4. **Approved:** The validated document is accepted by representatives of each party of stakeholders as an appropriate statement of requirements for the project. The developers then use the requirements document as a guide to implementation and to check the progress of the project as it develops.

1.2 How to Use This Document

We expect that this document will be used by people with different skill sets. This section explains which parts of this document should be reviewed by various types of readers.

Types of Reader

- GameMaster (DMs)
 - o Only Section 2 is necessary to read
- Players
 - o Only Section 2 is necessary to Read
- Future Developers/Content Creators
 - o Sections 2 And 3 Are most useful
- Project Overseers
 - o The entire document is useful

Technical Background Required

Section 2 is intended for a less technically savvy audience, which will include the general user, Players, or gamemasters.

Sections 3 and 4 are intended for more of a developer view, as technical CS terms or other project-specific language will be used

Overview Sections

Sections 1.3, 1.4, and 1.5 are very useful for a general overview of the product

Reader-Specific Sections

There are currently no sections designated for only the eyes of one group, but section 3 is more technical and is discouraged for those not interested in understanding the product at a deeper technical level

Section Order Dependencies

Currently, reading sections 1.3-1.5 will be a good place to start reading this product. 2.0 and 2.1 should logically follow. Section 3 is one of the last reads a user should do.

1.3 Scope of the Product

The scope of this project is to allow one superuser and multiple normal users to interact with a 3D space and manipulate various characters, or tokens, freely around this space. This space should be editable by the superuser, and they should also control which tokens the normal users can move around the space. As users request updates, changes should be visible to all users.

1.4 Business Case for the Product

In my experience playing TTRPGs like Dungeons and Dragons, combat is a vital role. In combat, positioning of both allies, enemies, and terrain is vital in making every combat encounter feel unique and developing strategies. One program that helped with this is called Foundry VTT.

However, Foundry is not without its flaws. Foundry is designed to work on a 2D plane only. While there are times when this is sufficient, it demonstrates deep flaws whenever either the enemies or the players start to think in three dimensions, whether this be from flight, or burrowing into the terrain. This can lead to confusion, as the map doesn't clearly mark these statuses. To make this complicated, according to the DND 2024 wikidot, 20% of the base 'species' a player can have the ability to fly. Thus, players will always have the option to fly, without items, spells, or other abilities. Thus, it is imperative to create a solution to reduce confusion at the table and streamline the TTRPG experience.

1.5 Overview of the Requirements Document

1. All normal clients are browser-based
2. This supports multiple normal users at once
3. All users can change perspective on the 3D space
4. All users can move tokens in 3D space
5. The Superuser (Or gamemaster) can control who can move certain tokens
6. The gamemaster can create/edit maps

2. General Description

This section will give the reader an overview of the project, including why it was conceived, what it will do when complete, and the types of people we expect will use it. We also list constraints that were faced during development and assumptions we made about how we would proceed.

2.1 Product Perspective

Hi! I'm Parker Hoffman, and I'm a Computer Science student who enjoys playing TTRPGs like Dungeons and Dragons. After playing D&D and using Foundry VTT, a map management software for D&D, I came into contact with one of its weaknesses. Since Foundry is designed on a 2D plane exclusively, it makes it difficult to imagine or quickly tell things about the depth involved in a scenario. The height of a room must be asked on entry (or necessity of the information) of a new room, which leads to repetitive questioning from some of the players to the GameMaster. In addition, the GameMaster already has a large amount of information they must constantly balance. Balancing the actual height/depth of a player/enemy can be difficult, and makes calculating projectile distance, range, or if the player/enemy is in the AoE of an action difficult. This can lead to some arguing over the table that could be easily avoided. As such, this product aims to fix this by providing a map management software with a 3-Dimensional approach from the start. As such, it will benefit both Players and the GameMasters by reducing confusion and streamlining the experience of playing a TTRPG together.

2.2 Product Functions

This product is designed to be the map for a TTRPG system. As such, it should be able to run in tandem during a session. In addition, it should be able to do the following:

- Create/Edit Maps
- Add Interactable Features to Maps
- Create/Edit Tokens
- Upload Images for Tokens
- Move Token in 3D space
- View/change perspective in 3D space
- Basic User Management

2.3 User Characteristics

The main group of people using this app are people with an interest in playing a TTRPG. These users typically desire a system that just works without any extra thought on their part. In addition, these users typically start with some basic computer system understanding. As such, they are expected to have very little technical training to use this application. Thus, it means that it is imperative to design an easy-to-pick up and intuitive system. The most complex thing anyone should be doing with my program is designing a map. Thus, it needs to be intuitive on how to design the map, build it, and place features. This is the largest obstacle for a user to overcome.

The only thing the users will need to be an expert in is how to play the TTRPG system they desire, which is outside of my program's scope.

2.4 General Constraints

- This program will rely on existing web-browser technology for Players to connect and interact with the system.
- This program will rely on existing internet protocols, including Sockets, to transfer information.
- The Player's client will be totally Browser-based, including the 3D environment. This also means it must be mobile responsive and fairly lightweight to support running on any browser.
- The main tech stack is a C#-based console app that relies on existing .NET frameworks to help handle the system.

2.5 Assumptions and Dependencies

For distribution purposes, it is assumed there is an intuitive website that offers a downloadable .exe file for the Gamemaster to run on their compatible device.

It is assumed the GameMaster will have a relatively recent computer with Windows Operating System installed and running on the device to host this program.

It is assumed that every user (GameMaster or Player) will have a consistent access to reliable internet throughout the entire period of use.

It is assumed that every end user will have a unique device with an up-to-date browser.

3. Specific Requirements

This section of the document lists specific requirements for GamemasterScrawl. Requirements are divided into the following sections:

1. User requirements. These are requirements written from the point of view of end users, usually expressed in narrative form.
2. System requirements. These are detailed specifications describing the functions the system must be capable of doing.
3. Interface requirements. These are requirements about the user interface, which may be expressed as a list, as a narrative, or as images of screen mock-ups.

3.1 User Requirements

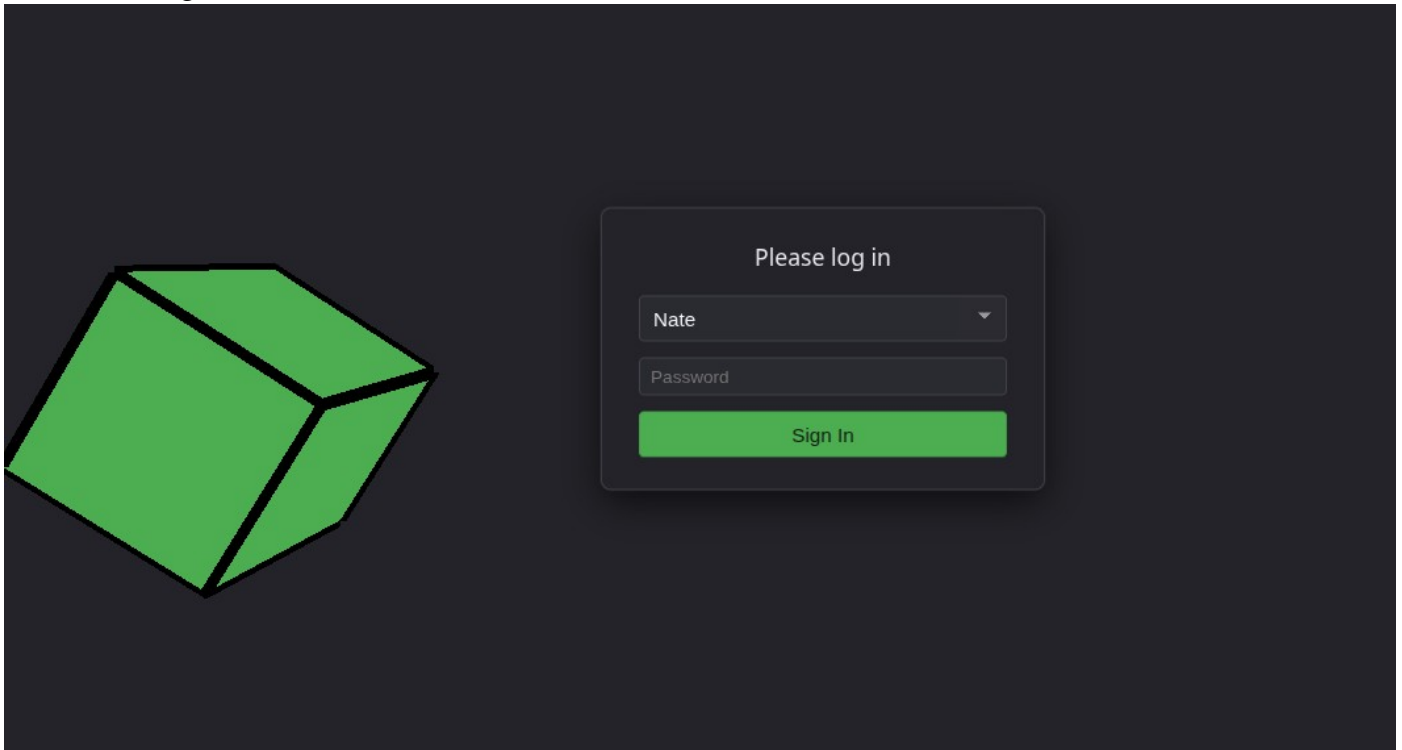
This system needs to be able to handle multiple users interacting and request changes on the data at once. It needs to visually display the data in an intuitive 3-dimensional view for ease of understanding. It needs to have intuitive ways to move the data

3.2 System Requirements

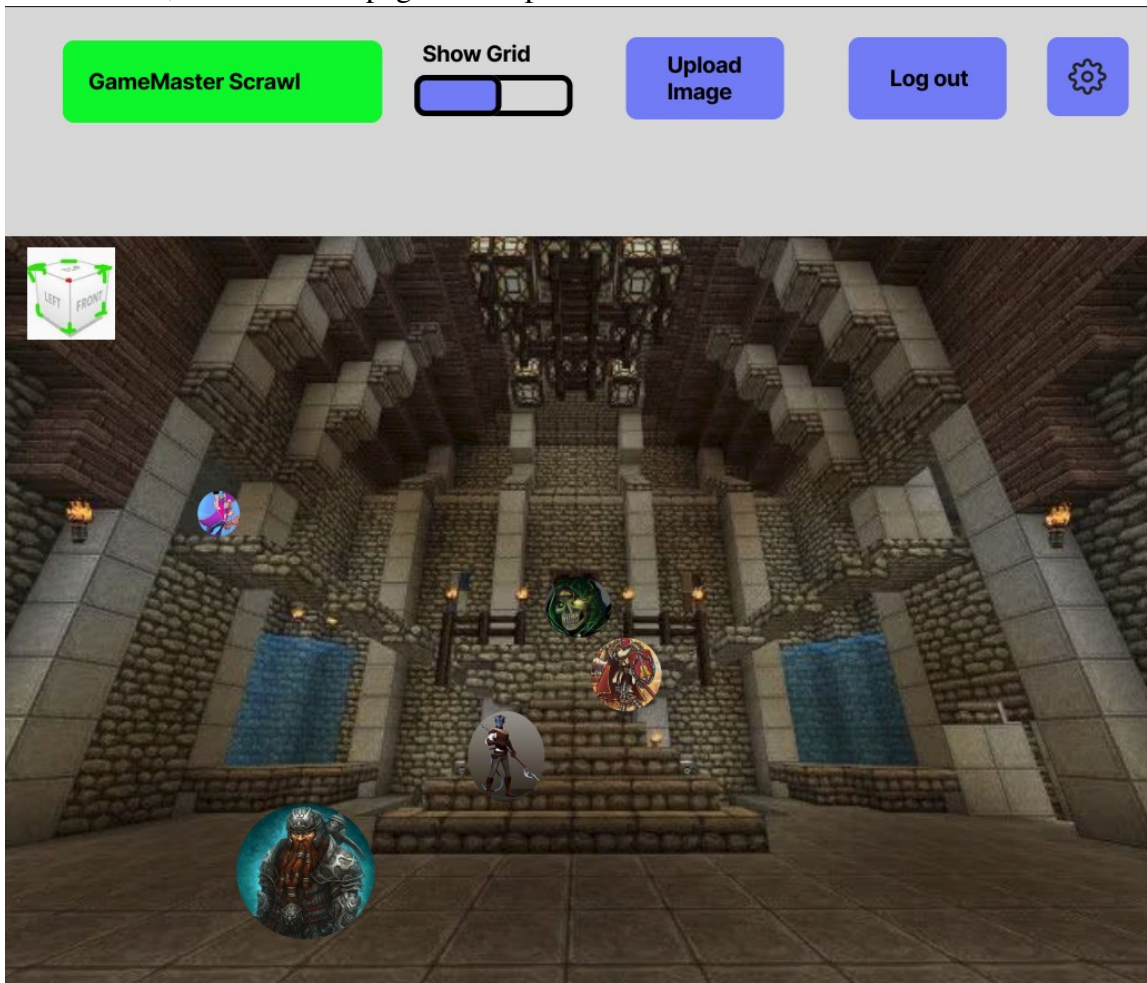
- Real-Time multi-user Token Management
- Internal Map Creation and Editing
- Token Permission per user
- 3D camera movement within a web browser environment
- Maintain a private data store for use and persistence after closing the application

3.3 Interface Requirements

First, the login screen:

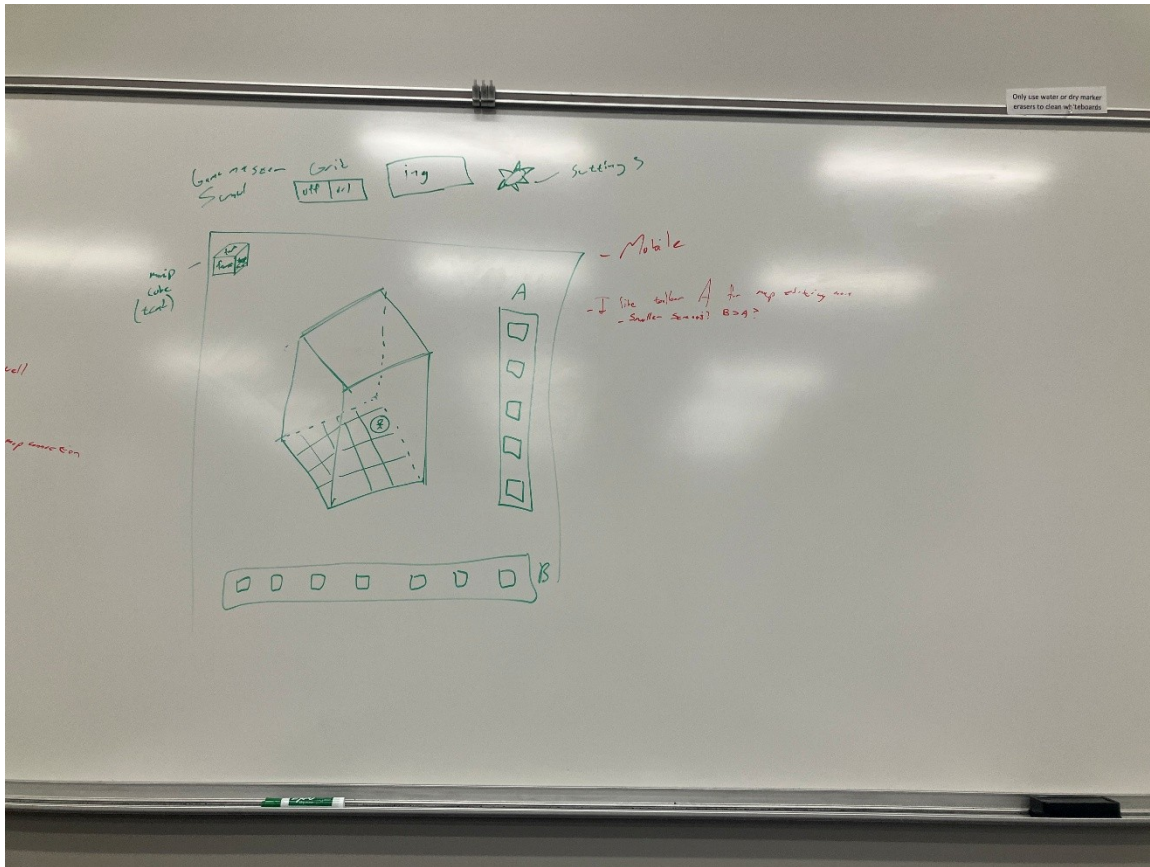


Then, the first Homepage Mockup:



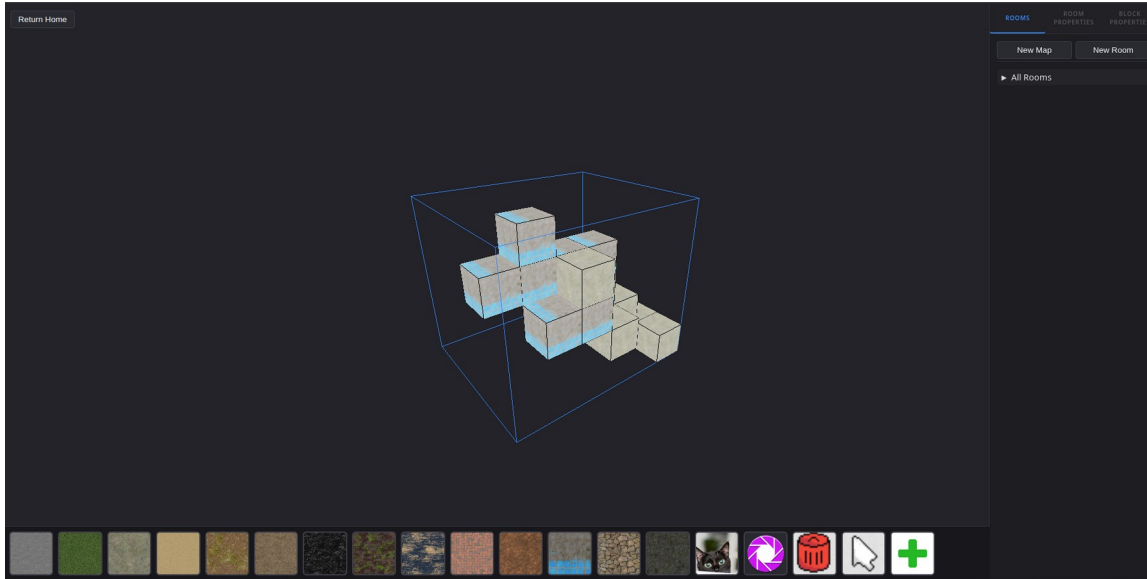
Above is the main interface screen for a player. The main hub at the top includes a way to show the grid, and the interactable spaces on the map. In the image, it is switched to the 'off' state.

- Next to that is the ability for a user to upload an image to be assigned to a token.
- Next to that, the user can sign out of this account
- Beneath the upper bar is the main map space. The user can rotate their camera by interacting with the cube in the upper-left part of the map. By moving their mouse around that, they can rotate the camera (and their perspective) around the current space
- To move a token, it should just be drag-and-drop

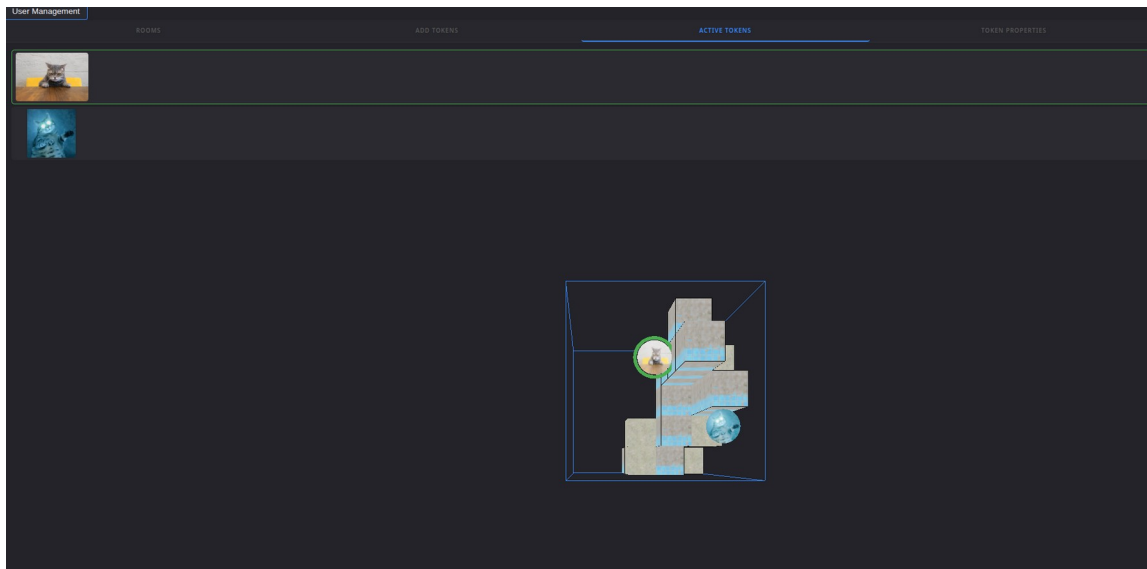


Above is an early draft of the UI, with sidebars to visualize.

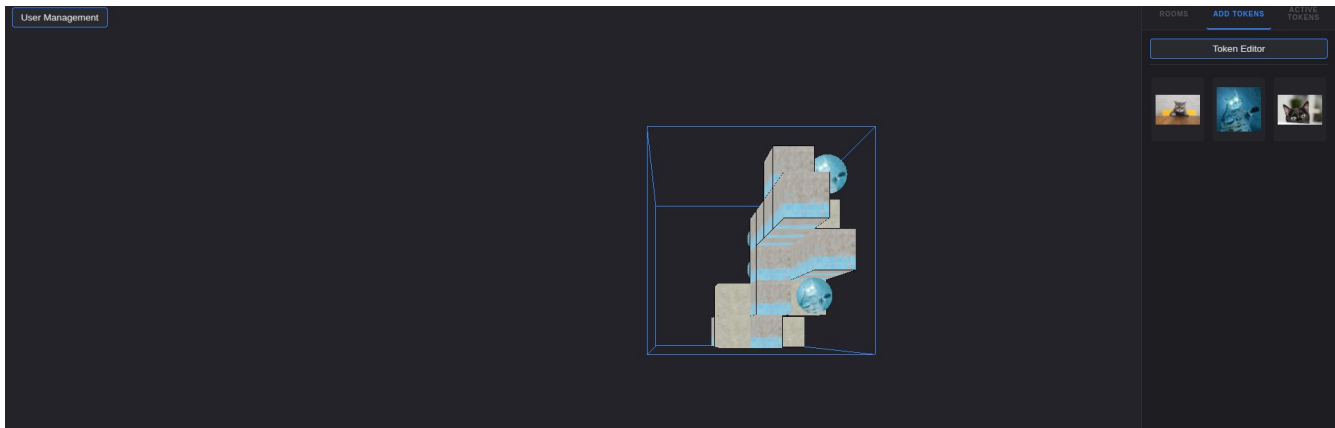
The addition screen has evolved to the following:



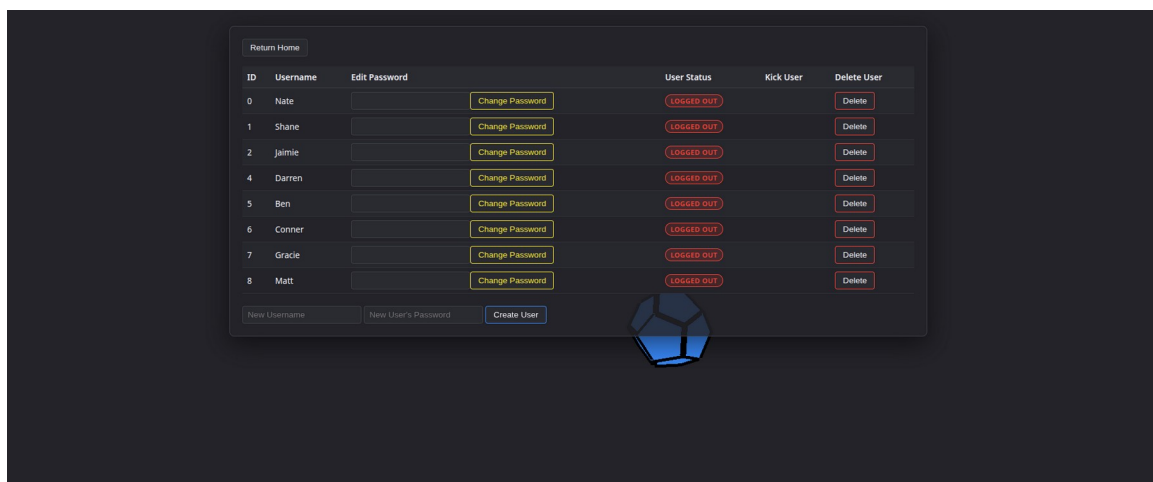
The above concept is also used on the home screen. Here is a screenshot before styling:



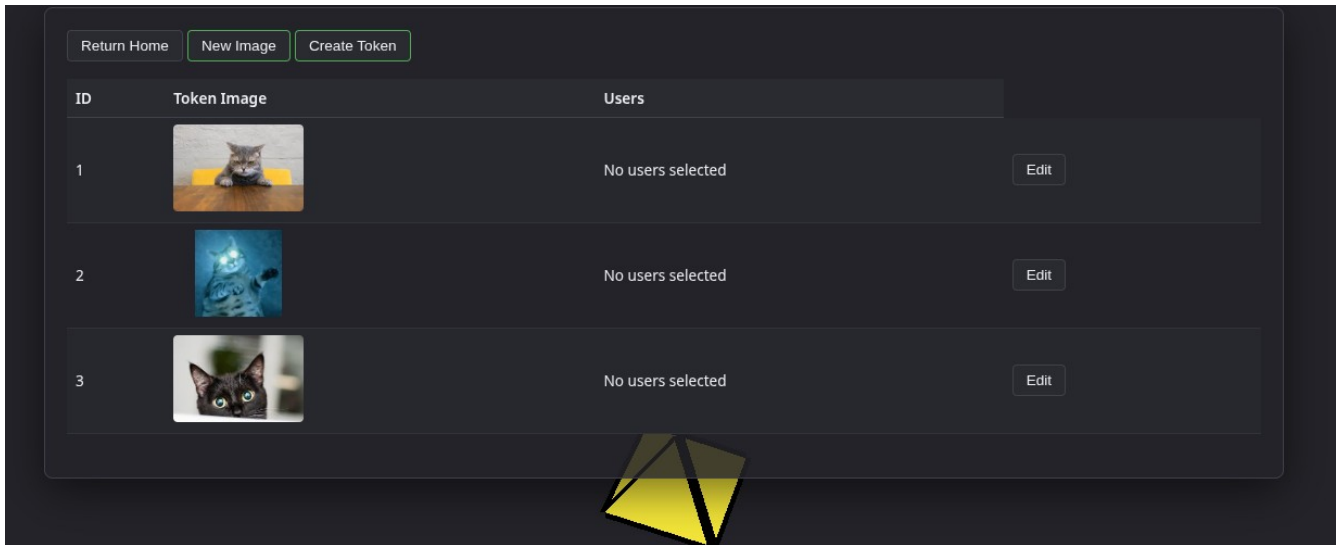
Here is a styled version of the home screen:



Next, there is a basic user management screen. This will populate with users as they are created, and will allow the host to edit the passwords and do basic user management



There is also a token editing screen. It follows a similar principle.



4. Appendices

Context Diagram for
Gamemaster Scrawl

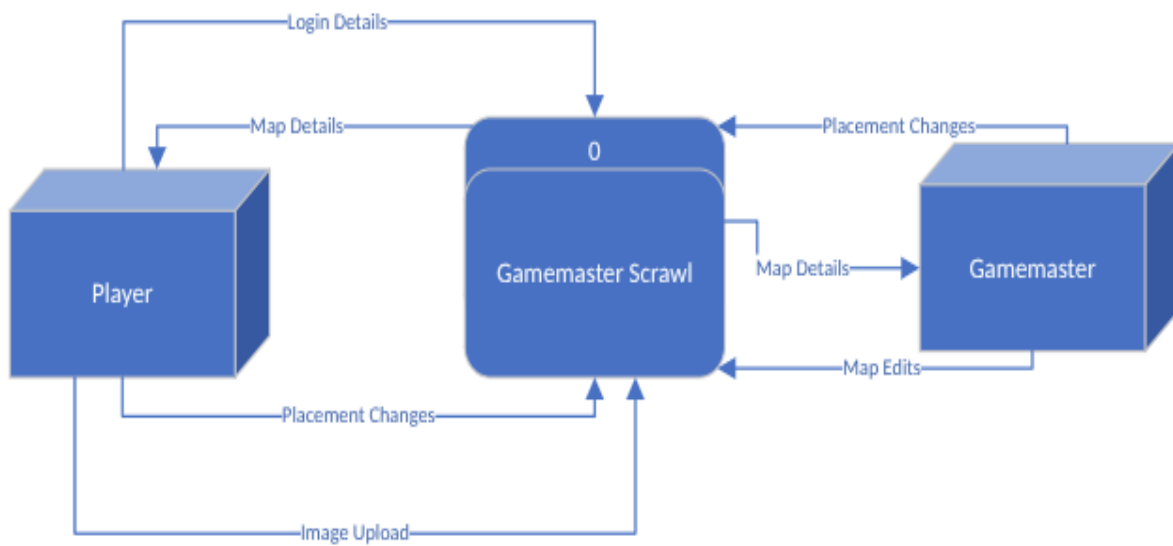
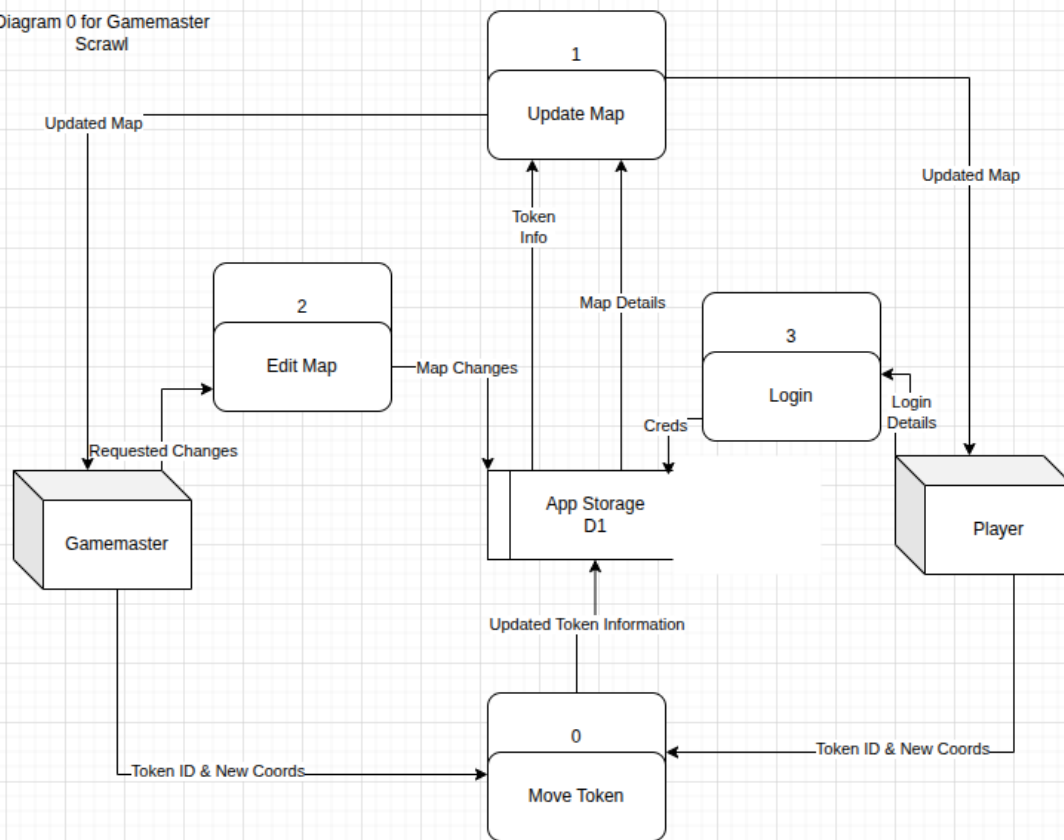
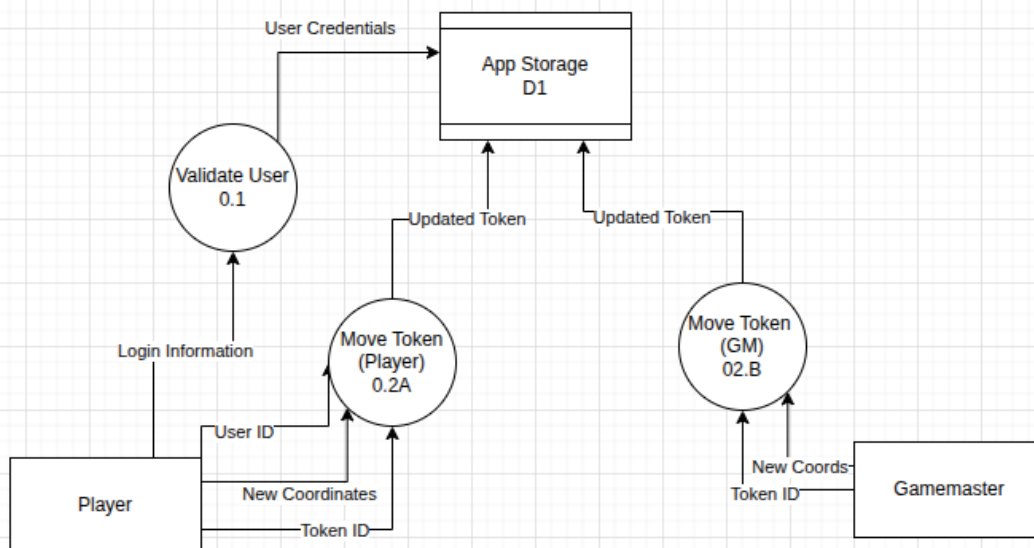


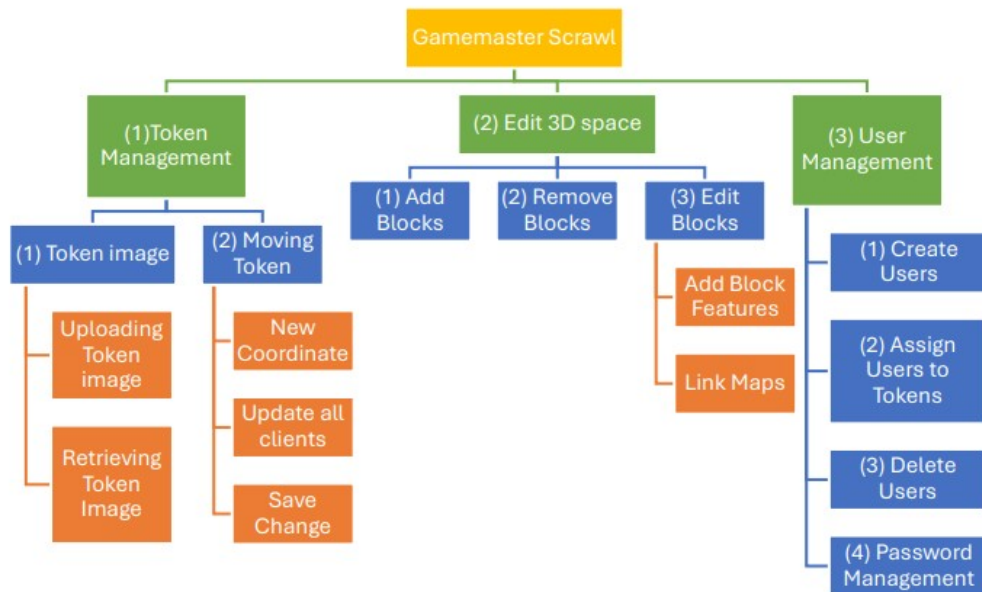
Diagram 0 for Gamemaster Scrawl



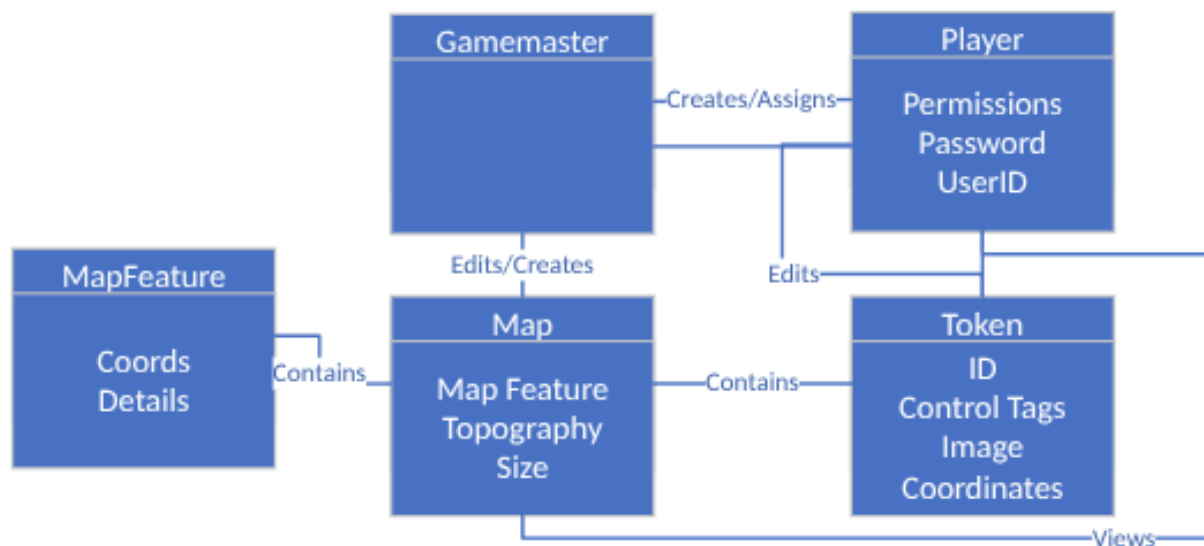
Gamemaster Scrawl: DFD
Move Token: 0

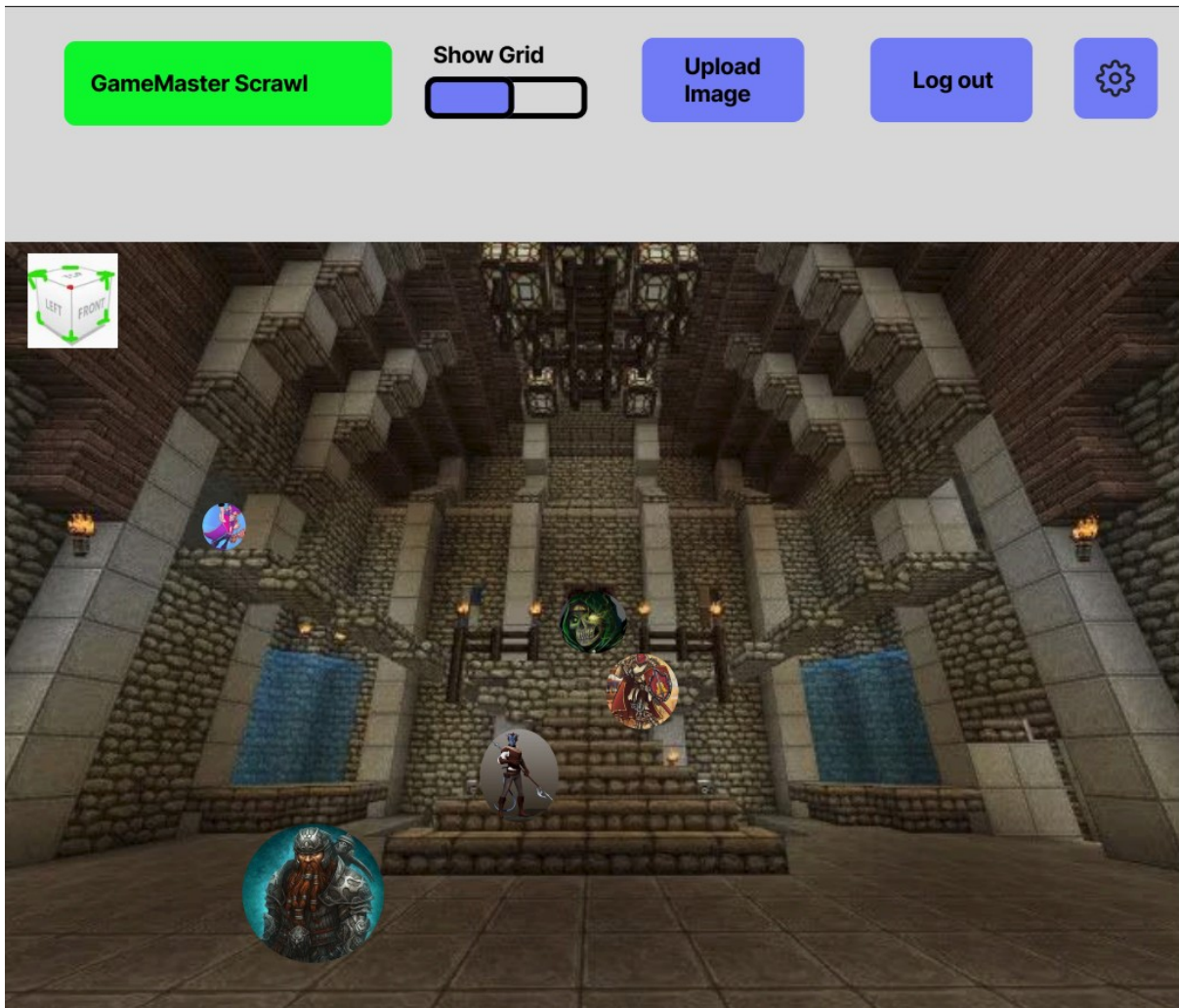


Functional Decomposition Diagram:

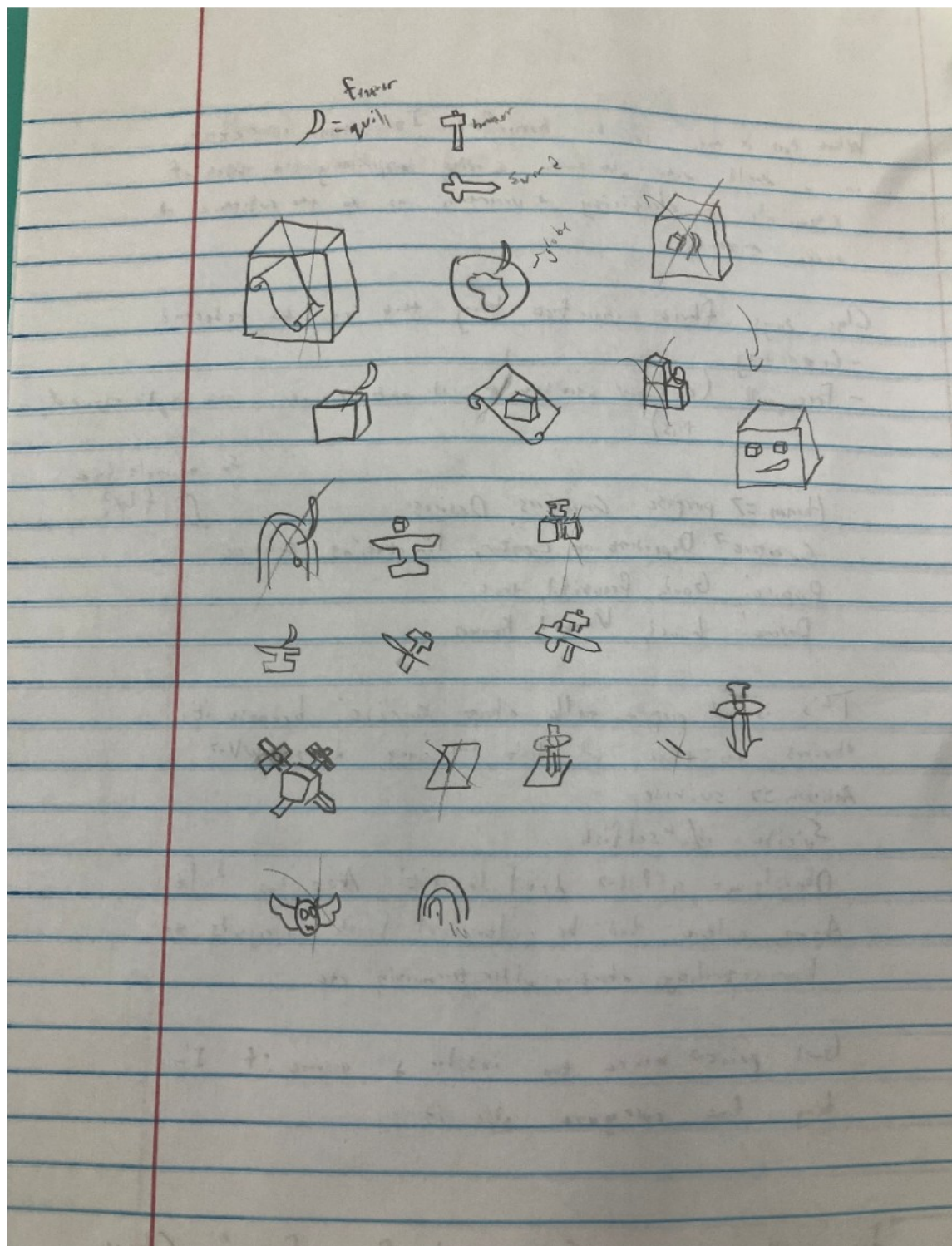


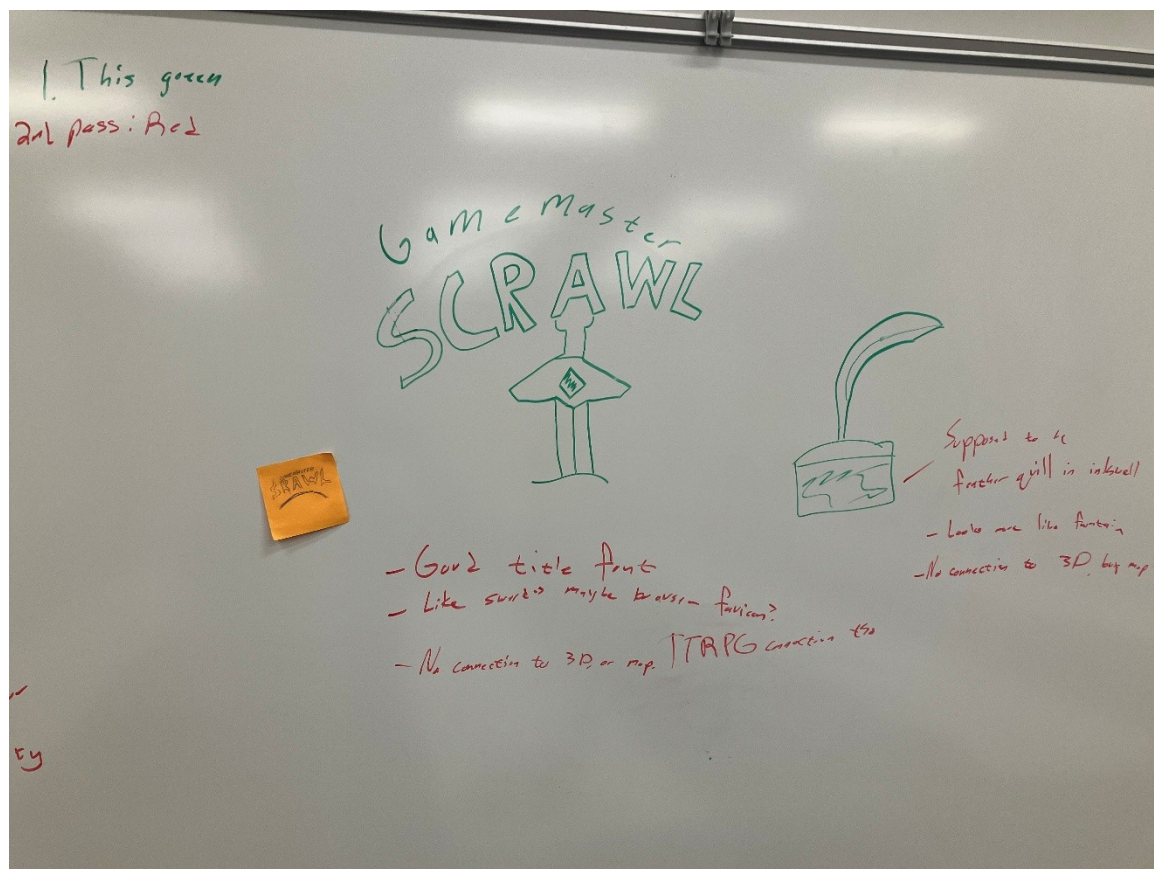
Object Model

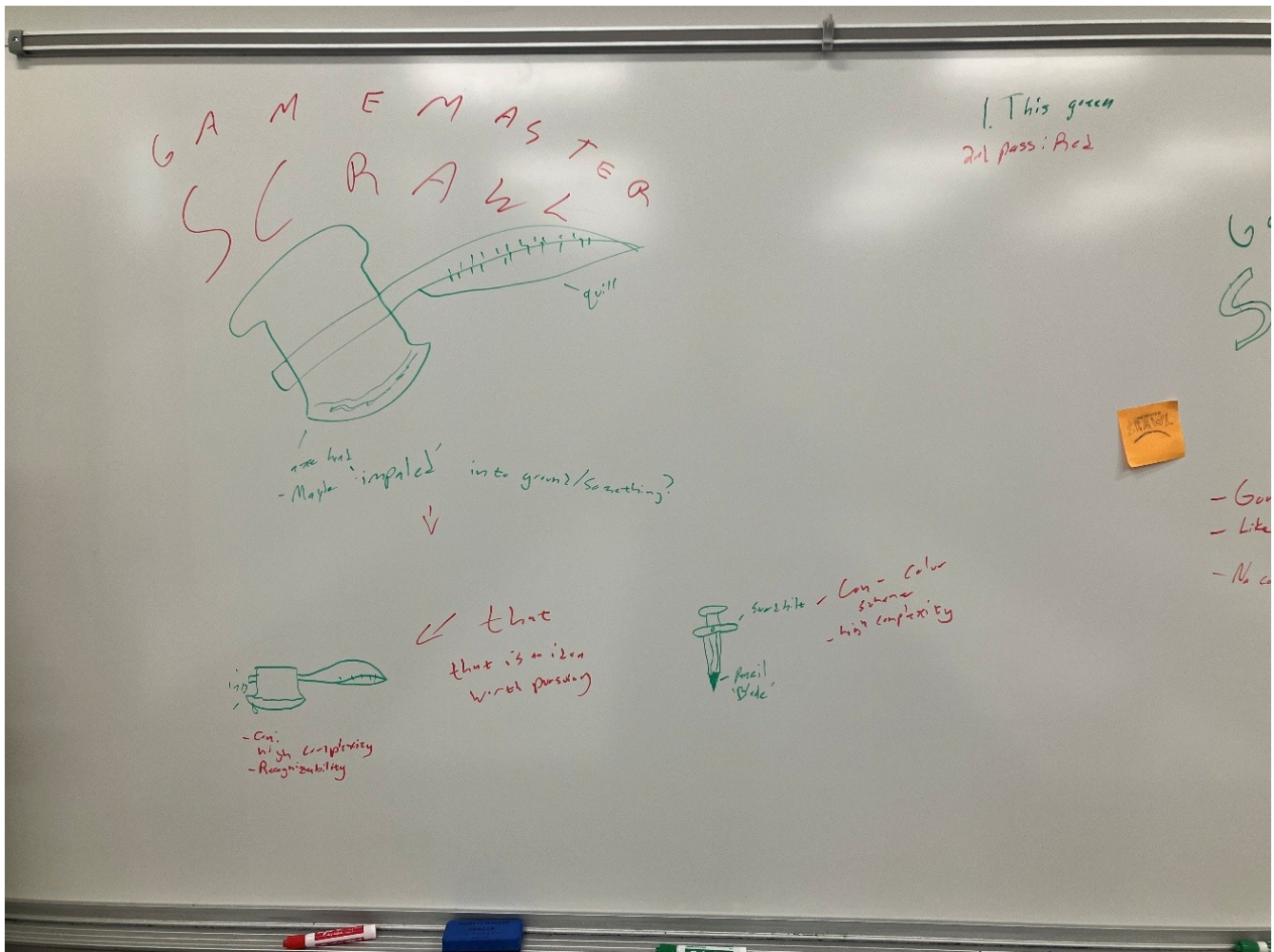




Below is a list of Logo Drafts



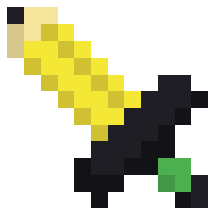




Below is the actual Logo:



And the Favicon:



Wish-listed Features:

- Map Import Utility
- Assigning sub-Superusers
- Measurement Tools (AoE bubbles, Distances)
- Animations
- Custom Model support
- 2D plane View(s)
- Dice rolling utility
- Turn Order Manager
- Built-In Stat-block manager

- Physics System
- Interactable Blocks

5. Glossary

TTRPG: This stands for Table-Top Roleplaying Game. This is when a group of people get together and play a structured version of make-believe over a set period of time. Different people at the table have different roles, like player or Gamemaster. The structure comes from a rule set they agree to play with. These rule sets would typically be referred to by names like Dungeons and Dragons, Pathfinder, Lancer, etc.

Gamemaster: The super user running the TTRPG session. This person has planned the entire adventure and is the main storyteller of the session. Sometimes referred to as 'Dungeon Master' or other terms, depending on the context

Player: This is any other person interacting with the story. These people make decisions that greatly influence the story

AoE: This stands for Area of Effect. This is the space a action takes up. Think if a loud sound were to play, its area of effect would be everywhere it can be heard. These typically take a spherelike appearance around the terrain. This is typically used in things like explosions.

Socket: This is when a client maintains a connection with a host (typically a server) over the internet, instead of terminating the connection after it receives the information it wants

Client: User interacting with the program. This will typically denote someone with no special permissions or access to features

Superuser: This is a user with higher access to various parts or features of the application

Map: This is the 3D space in which Tokens, or other actors are being manipulated within

Block: This is one grid space of the map. This can have interactable properties, be empty, or have the space filled

Token: This is a little circle (possibly with an image) representing an actor in the space. It could represent a player's character, an enemy, an ally, or any other moveable creature in the space. This is what is moved around in the space

6. References

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Project Schedule

#	Task Name	Start	Finish	Duration
1	Problem Statement Document	01/15/2026	01/15/2026	0 days
2	PDD Draft	01/15/2026	01/15/2026	0 days
3	Stand-Up Practice	01/29/2026	01/29/2026	0 days
4	Revise PDD	01/15/2026	01/15/2026	1 day
5	PDD Questions/Responses	01/15/2026	01/27/2026	13 days
6	PDD Final	01/15/2026	01/29/2026	15 days
7	Work Breakdown Structure	01/15/2026	02/05/2026	22 days
8	Project Schedule V1	01/15/2026	02/10/2026	27 days
9	Elevator Pitch Lab	01/15/2026	02/12/2026	29 days
10	Resume	01/15/2026	03/26/2026	71 days
11	Logo	01/15/2026	03/31/2026	76 days
12	TRD V2	01/15/2026	04/02/2026	78 days
13	Executive Summary	01/15/2026	04/21/2026	97 days
14	TRD V3	01/15/2026	04/21/2026	97 days
15	User Manual	01/15/2026	04/23/2026	99 days
16	Documentation	01/15/2026	04/23/2026	99 days
17	Basic Backend Server Setup	01/15/2026	01/15/2026	1 day
18	Socket Connections	01/15/2026	01/15/2026	1 day
19	Implement 3D Viewer	01/15/2026	01/15/2026	1 day
20	Implement Persistent Storage	01/15/2026	01/15/2026	1 day
21	Actual Webpage Visible	01/15/2026	01/15/2026	1 day
22	Share Github Repository	01/14/2026	01/14/2026	0 days
23	User Management	01/15/2026	01/15/2026	1 day
24	Investigate 1-Page Application	01/15/2026	01/15/2026	1 day
25	Host Bypasses Login	01/15/2026	01/15/2026	1 day
26	Open Webpage on App Run	01/15/2026	01/15/2026	1 day
27	Close Server Under Certain Conditions	01/15/2026	01/15/2026	1 day
28	Host Registration	01/15/2026	01/15/2026	1 day
29	Make Initial Screen Mockups	01/15/2026	01/15/2026	1 day
30	Make Initial Logo	01/15/2026	01/15/2026	1 day

#	Task Name	Start	Finish	Duration
32	Implement Map Editor	01/15/2026	01/15/2026	1 day
33	Implement Token Management	01/15/2026	01/15/2026	1 day
34	Create Realistic Scenarios for Testing	01/15/2026	01/15/2026	1 day
35	Run Testing Scenarios	01/15/2026	04/17/2026	93 days
36	Implement Room Saving	01/15/2026	01/15/2026	1 day
37	Adding Blocks	01/15/2026	01/15/2026	1 day

Project Status Report

Overall Status:  On Track

GamemasterScrawl (5) Parker Hoffman

Date: 2/10/2026

Status Code Legend

-  On Track: Project is on schedule
-  High Risk: At risk, with a high risk of going off track
-  At Risk: Milestones missed but date intact
-  Off Track: Date will be missed if action not taken

The project is ON TRACK the week of 1/26/2026 – 2/6/2026, due to the following:	 Due to setting up GitHub, and getting a basic server running with some of the required features finished, as listed below  Basic Hosting on a machine-to-machine basis  Socket Connection Infrastructure successfully running  Host registration complete  Page structure layout planning begun  Logo Design Started
Issues:	 Figuring out single-page application necessity/implementation (Or investigating socket connections in the alternative)
Milestones accomplished on the week(s) of 1/26/2026 – 2/6/2026:	 Hosting Solved  Socket Connection Started  Handled Registering the Host
Milestones planned this week, but not achieved with variance:	 Host Bypasses Login Screen  Basic User Management
Milestones planned for next week:	 Basic User Management  Host Bypasses Login  User Login System (Tokening of a sort)
Areas/questions for discussion:	Currently, nothing of note



Project Status Report

Overall Status:  On Track

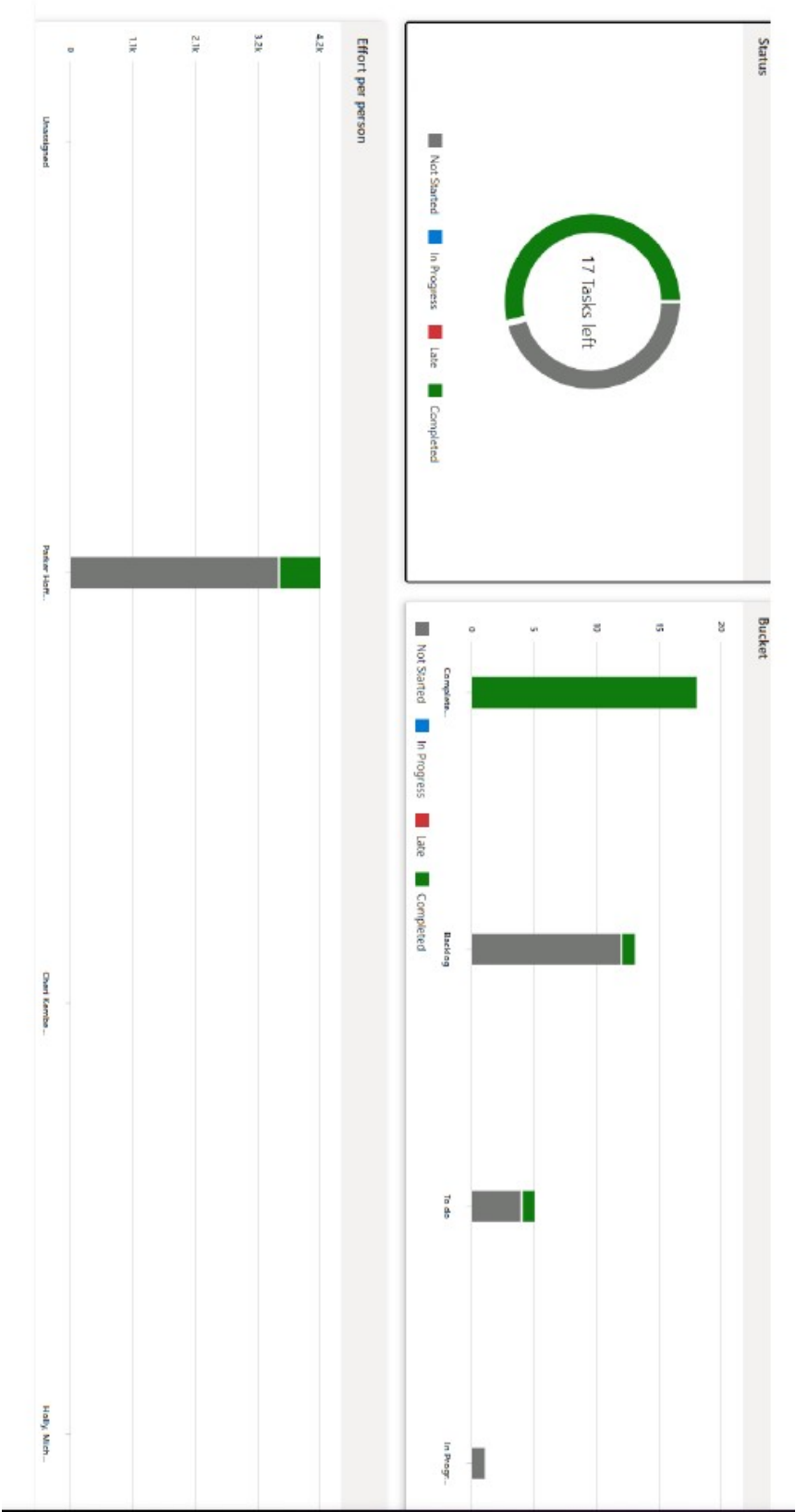
GamemasterScrawl (5) Parker Hoffman

Date: 2/23/2026

Status Code Legend

-  On Track: Project is on schedule
-  High Risk: At risk, with a high risk of going off track
-  At Risk: Milestones missed but date intact
-  Off Track: Date will be missed if action not taken

The project is ON TRACK the week of 2/10/2026 – 2/23/2026, due to the following:	 Started 3D manipulation  Basic Object creation  Designed Map Editor Screen
Issues:	
Milestones accomplished on the week(s) of 1/26/2026 – 2/6/2026:	 Initial implementation of 3D  Styling Established
Milestones planned this week, but not achieved with variance:	 User can place blocks  Map Persistence across runtimes
Milestones planned for next week:	 User can place blocks  Map Persistence across runtimes
Areas/questions for discussion:	Currently, nothing of note



Project Status Report

Overall Status: 🟢 On Track

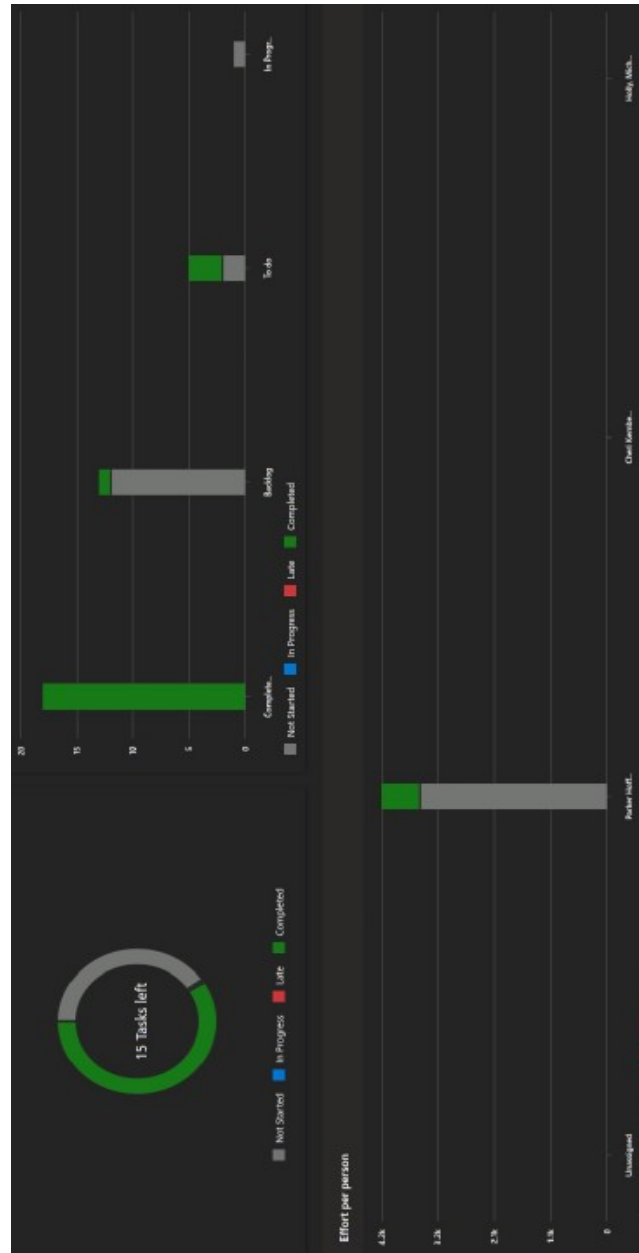
GamemasterScrawl (5) Parker Hoffman

Date: 3/8/2026

Status Code Legend

- On Track: Project is on schedule
- High Risk: At risk, with a high risk of going off track
- At Risk: Milestones missed but date intact
- Off Track: Date will be missed if action not taken

The project is ON TRACK the week of 2/23/2026 – 3/8/2026, due to the following:	● Started 3D Map Editing ● Correctly Renders room based off of persistent storage ● Material selection and block placement
Issues:	● Not satisfied with Block additions
Milestones accomplished on the week(s) of 2/23/2026 – 3/8/2026:	● 3D block placement ● Material Selection ● Changes saved ● Cleaned up 3D rendering
Milestones planned this week, but not achieved with variance:	● Feature placements (teleporters, room ties, etc.) ● Material Uploads
Milestones planned for next week:	● Refine 3D interactions ● Material Uploading ● Some basic Feature Blocks ● Begin Tokens
Areas/questions for	Currently, nothing of note



Project Status Report

Overall Status: 🟢 On Track

GamemasterScrawl (5) Parker Hoffman

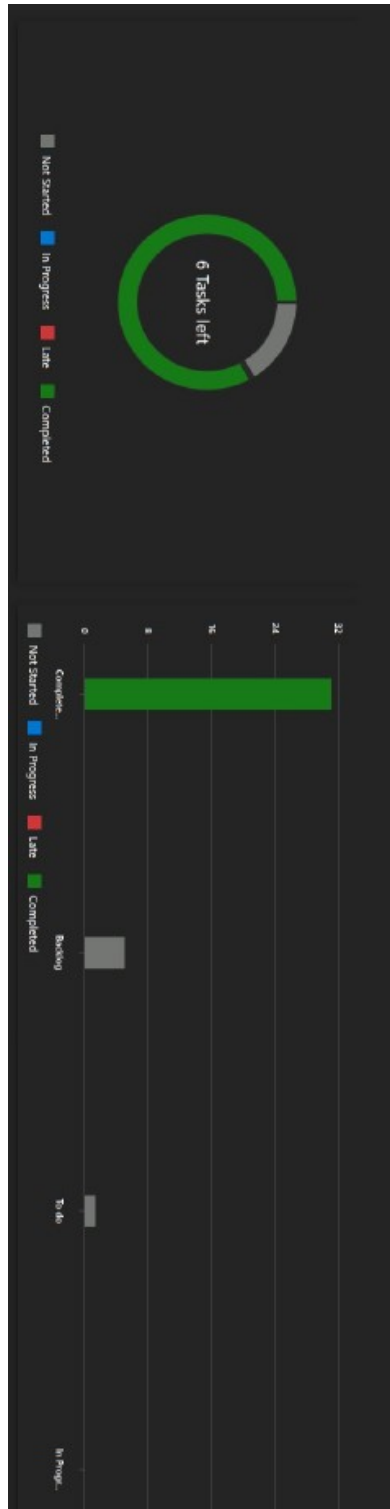
Date: 3/22/2026

Status Code Legend

- On Track: Project is on schedule
- High Risk: At risk, with a high risk of going off track
- At Risk: Milestones missed but date intact
- Off Track: Date will be missed if action not taken

The project is ON TRACK the week of 3/9/2026 – 3/22/2026, due to the following:	● Refined 3D Map Editing ● Correctly Renders room based off of persistent storage ● Material Upload
Issues:	● Time
Milestones accomplished on the week(s) of 3/9/2026 – 3/22/2026:	● Refined 3D block placement ● Material Upload ● Added a teleporter object
Milestones planned this week, but not achieved with variance:	● Feature placements
Milestones planned for next week:	● Batch Map editing ● Block/Room Editing Panes ● Begin Tokens
Areas/questions for discussion:	Currently, nothing of note





Project Status Report

Overall Status:  On Track

GamemasterScrawl (5) Parker Hoffman

Date: 4/5/2026

Status Code Legend

-  On Track: Project is on schedule
-  High Risk: At risk, with a high risk of going off track
-  At Risk: Milestones missed but date intact
-  Off Track: Date will be missed if action not taken

The project is **ON TRACK** the week of 3/22/2026 – 4/5/2026, due to the following:

-  Logo/Favicon completed
-  Token Creation Working

Issues:

-  Time

Milestones accomplished on the week(s) of 1/26/2026 – 2/6/2026:

-  Logo/Favicon completed
-  Token Creation Working

Milestones planned this week, but not achieved with variance:

-  Token Rendering
-  Token Addition to map

Milestones planned for next week:

-  Token Addition to map
-  Token Rendering
-  Token Selection
-  Token Manipulation

Areas/questions for discussion:

-  Letter of recommendation

Project Status Report

Overall Status:  **On Track**

GamemasterScrawl (5) Parker Hoffman

Date: 4/29/2026

Status Code Legend

-  On Track: Project is on schedule
-  High Risk: At risk, with a high risk of going off track
-  At Risk: Milestones missed but date intact
-  Off Track: Date will be missed if action not taken

The project is ON TRACK the week of 3/22/2026 – 4/5/2026, due to the following:	 App Completed
Issues:	N/A
Milestones accomplished on the week(s) of 1/26/2026 – 2/6/2026:	 Fixed Map Creation Bug  Commenting  Home Screen Complete
Milestones planned this week, but not achieved with variance:	N/A
Milestones planned for next week:	 Presentation
Areas/questions for discussion:	

GamemasterScrawl

User's Guide

A guide to simplifying TTRPG Map Management

Parker Hoffman
4-22-2026

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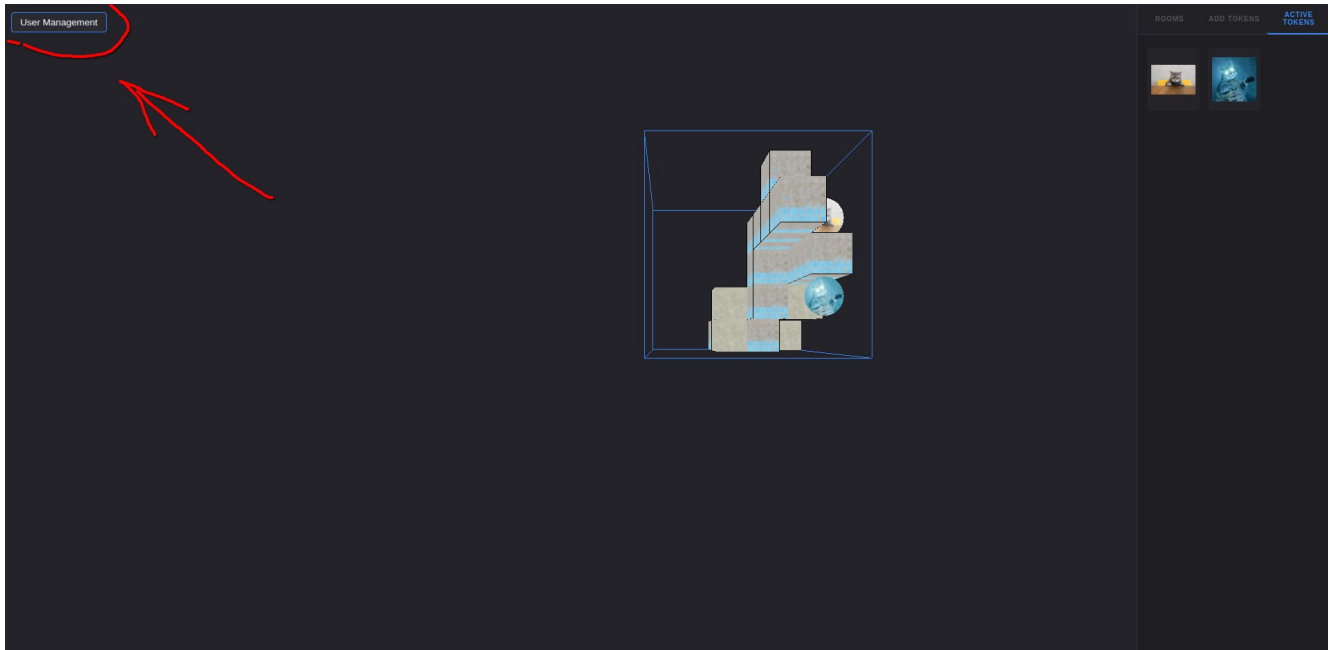
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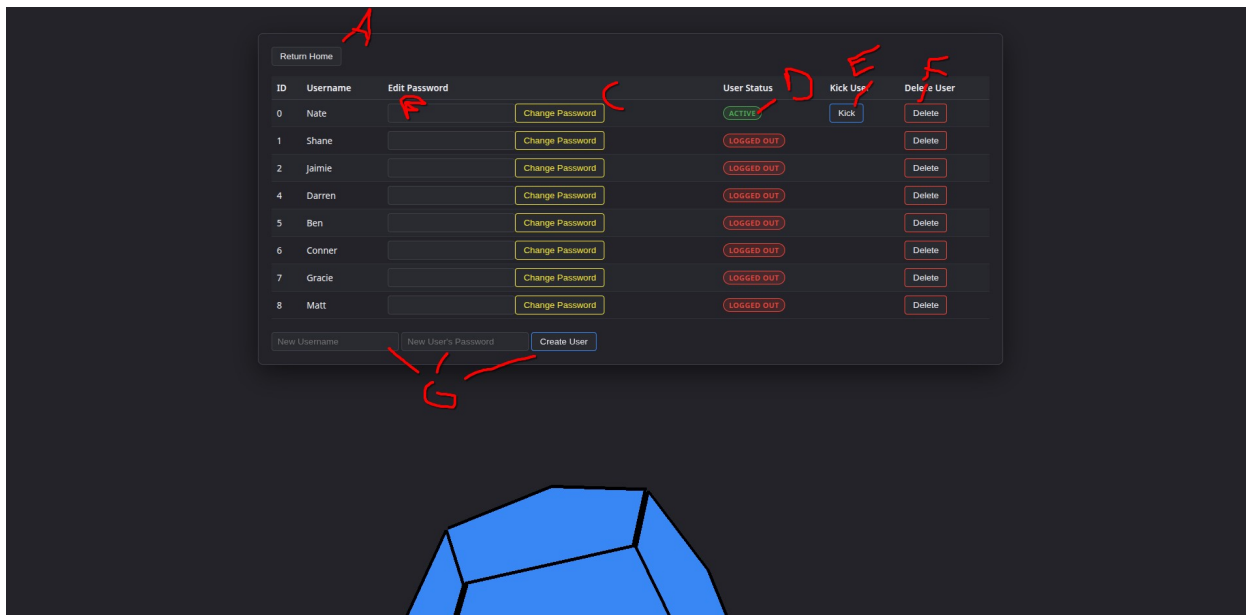
4.4.3: Moving a token (11)

1: User Management

In order to allow other users to sign in, accounts for all non-host users must be created. To do that, the host must first click the button labeled 'User Management' in the top left corner of the home screen. See the image below to see what your screen might look like:



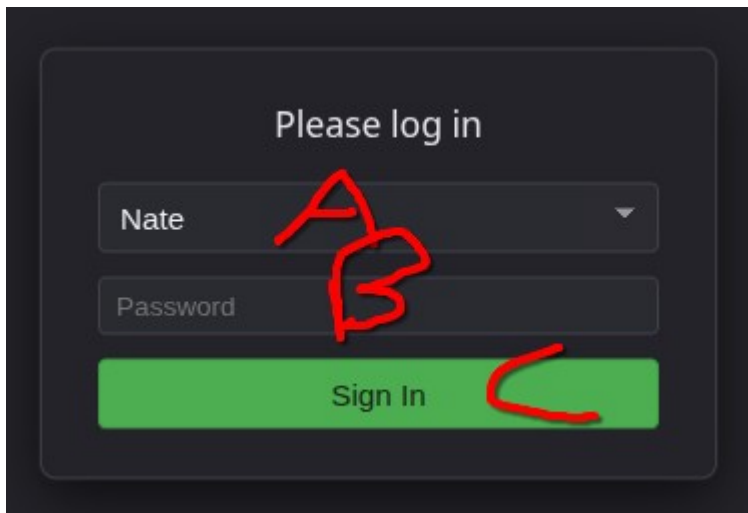
You will then see this screen.



- A. This button will return the user to the home screen
- B. : this is where you can input a different password for the use this row represents. See C for where to submit it. Case is important!
- C. This button will submit and change the password for the user permanently. Be careful!
- D. This will indicate if a user is currently logged in under this account
- E. This button will only appear if a user is signed in under this account. Upon clicking of this button, it will log that user out and send them back to the log in screen
- F. Upon clicking this button, this account will be deleted. Be careful: This cannot be undone!
- G. This is where you create a new user. Text must be entered into the first field, this is the username from the user. *This cannot be changed after submission.* The second field is the password for this account. The third is how you submit and create the new accounts

1.1: User Login

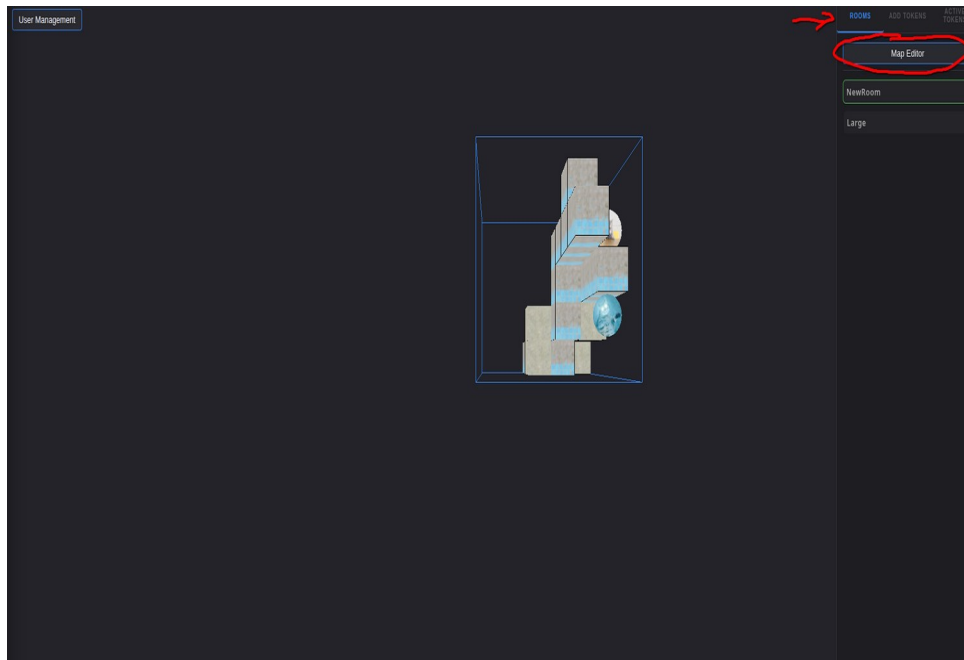
If the user is not the host, they will see the following screen:

The image shows a login interface on a dark background. At the top, the text "Please log in" is centered. Below it are three elements: a dropdown menu with the text "Nate" and a downward arrow, a text input field with the placeholder "Password", and a green rectangular button with the text "Sign In". Red handwritten letters are placed next to each element: a red "A" next to the dropdown, a red "B" next to the password field, and a red "C" next to the "Sign In" button.

- A: This is a dropdown. The user can choose the name of the account they want to log in under
- B. This is the input for the password. As the user, enter in the correct password for the user selected
- C. Click this to attempt to log in as the selected user with the given password, If the password is not correct, you will be notified. Otherwise, you will be sent to the home screen.

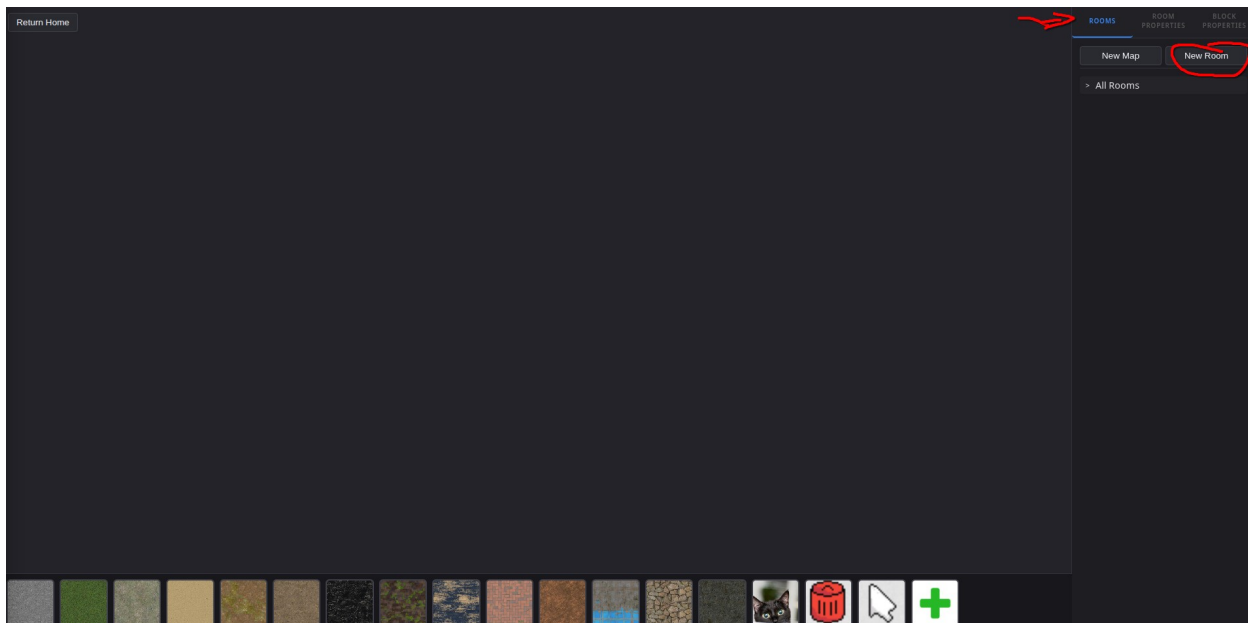
2. Map Creation

In order to have a map to deal with, the map must first be created. As the host, open the home screen. First, as the host, in the pane on the right, select the 'Rooms' tab. Then, click the 'Map Editor' button.



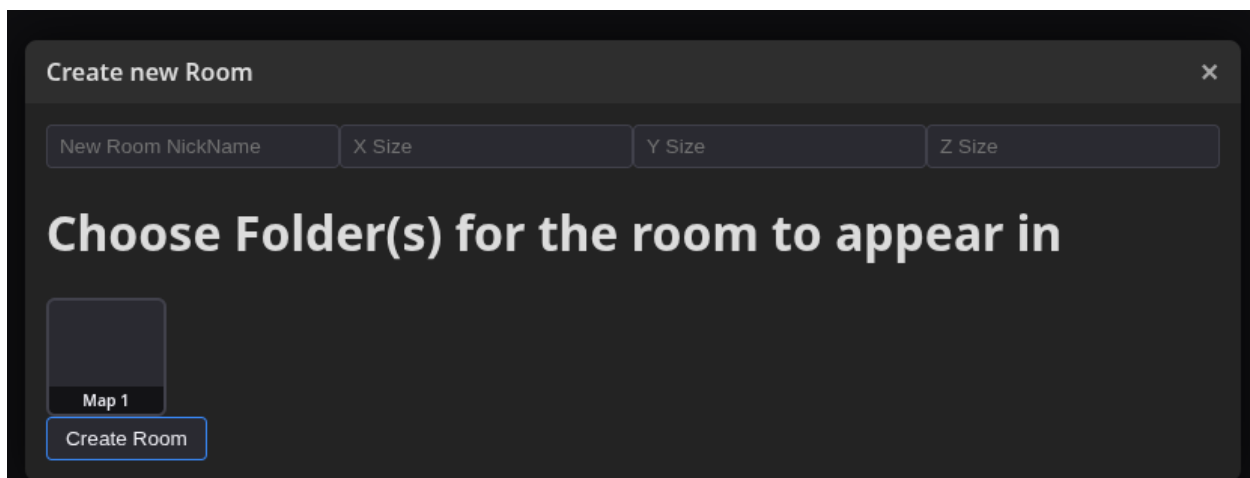
To select a room to edit it, in the pane on the right, choose the 'Rooms' tab. Then, choose the 'map' the room has been assigned, or you can find it under the 'All Rooms' section. Click on the room you want to edit, and it will appear in the editing pane

2.1: Creating a Room



To create a new room, have the host open the map management screen. Then, in the right sidebar, select the 'Rooms' tab, and click the 'new Room' button

A new window will now be popped open for creating a fresh map. It contains a nickname for the room, the dimensions for the room, and optional selectors for what 'maps' (or folders' to store the map in. Click the 'Create Room' button after inputting the nickname and dimensions to create the room. It will



be auto-selected

2.2: Editing a map

After selecting a room, it will appear in the main pane. In the bottom of the screen, there is the following toolbar:

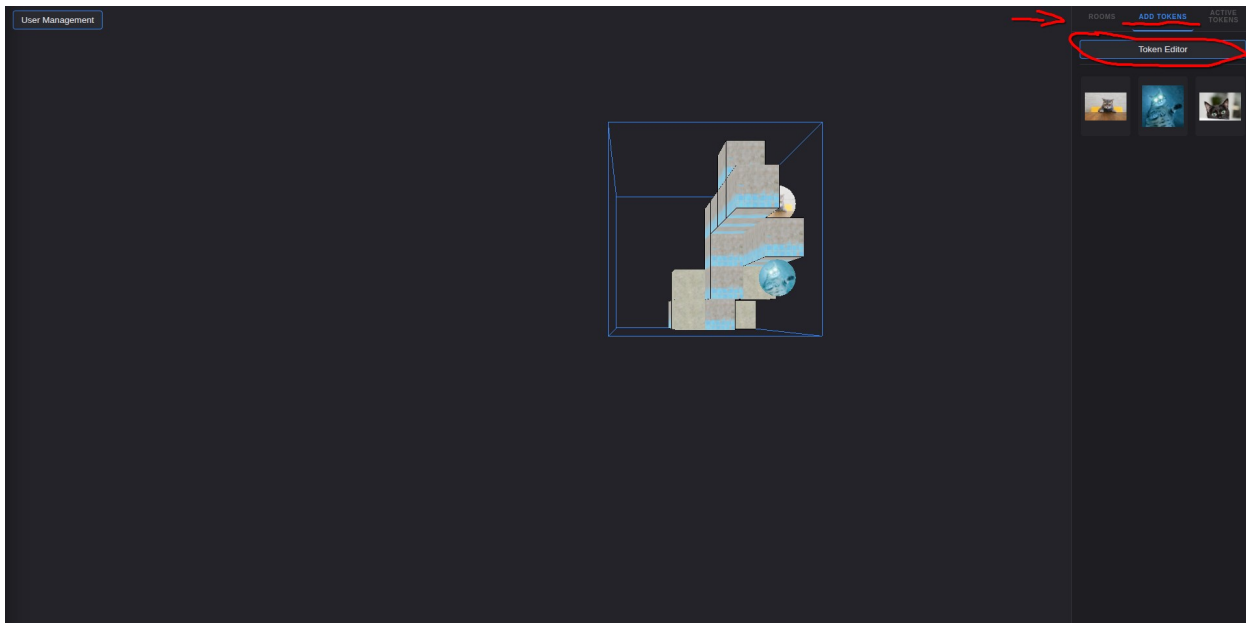


- A. These are all the blocks that can be added. Just click a block to select it, and then click on the map to add a block with that texture there
- B. This is the ‘delete block’ option. Click to select it, and then when clicking on a block on the map, it will delete it.
- C. This is the ‘select block’ option. It allows selection of a specific block for editing other properties
- D. This button allows you to upload any image to use as a texture for a block. Clicking it will open file explorer for file selection.

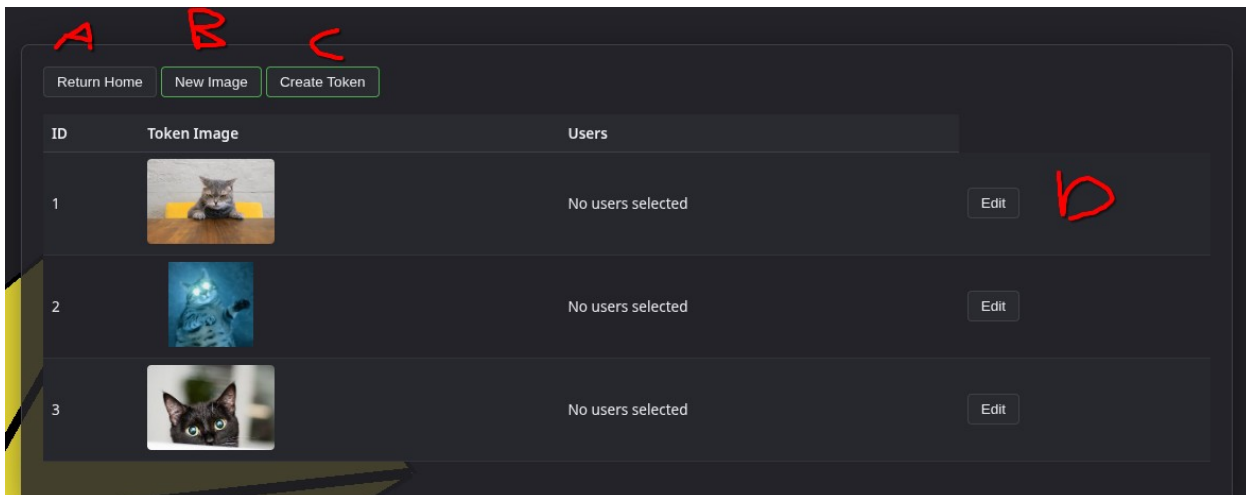
The map will automatically save any changes made to it.

3. Token Management

As the host, on the home screen, select the ‘Add Tokens’ Tab in the right sidebar. Then, click the ‘Token Editor’ button to navigate to the Token Management screen.



Once on this screen, you will see the following:

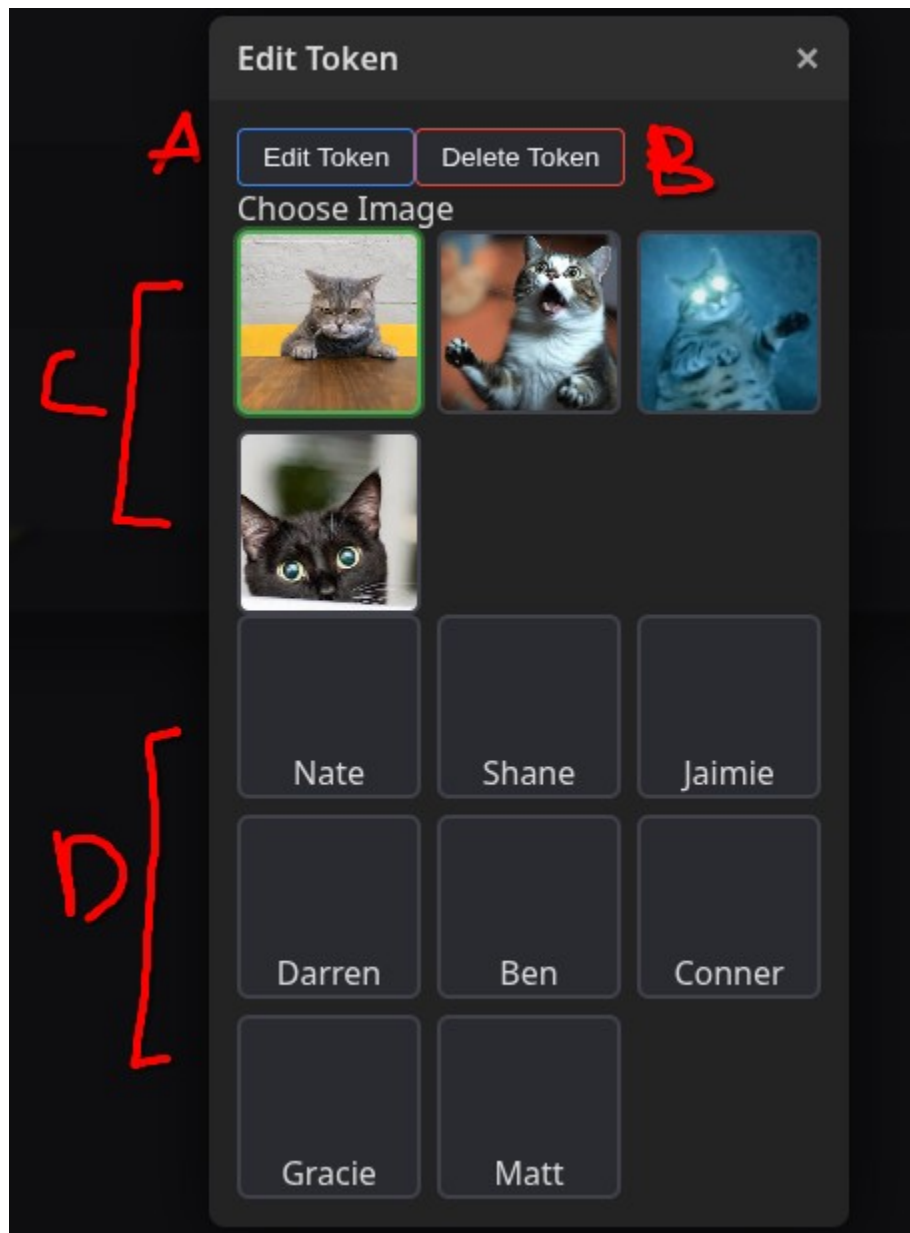


A. This button will take the user back to the Home screen

B. This button will allow the user to upload a new image that can be used for tokens. Clicking it will open a File Explorer window for file selection

C. Clicking this button will open a dialog to create a fresh token. The dialog is the next diagram

D. Clicking this button allows for editing of the existing token in this row. It opens the dialog in the next diagram



A. This is the button that saves changes to the token. If you are instead creating a token, it saves and creates the new token.

B. This button will only show up if you are editing a token. On click, it deletes the token. This cannot be undone

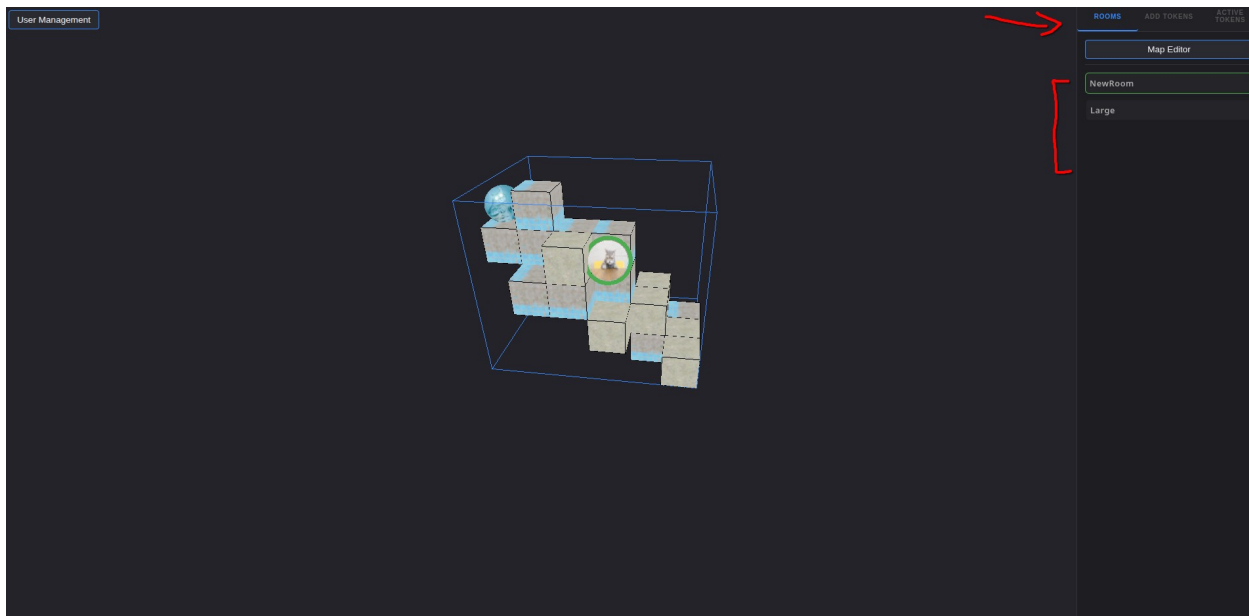
C. This is the image for the token. Only 1 image can be selected. The only options are images you have previously uploaded

D. This is a list of users who can move this token. It reads from the user list set up from section 1. The host is always included for any token.

4. Map Manipulation

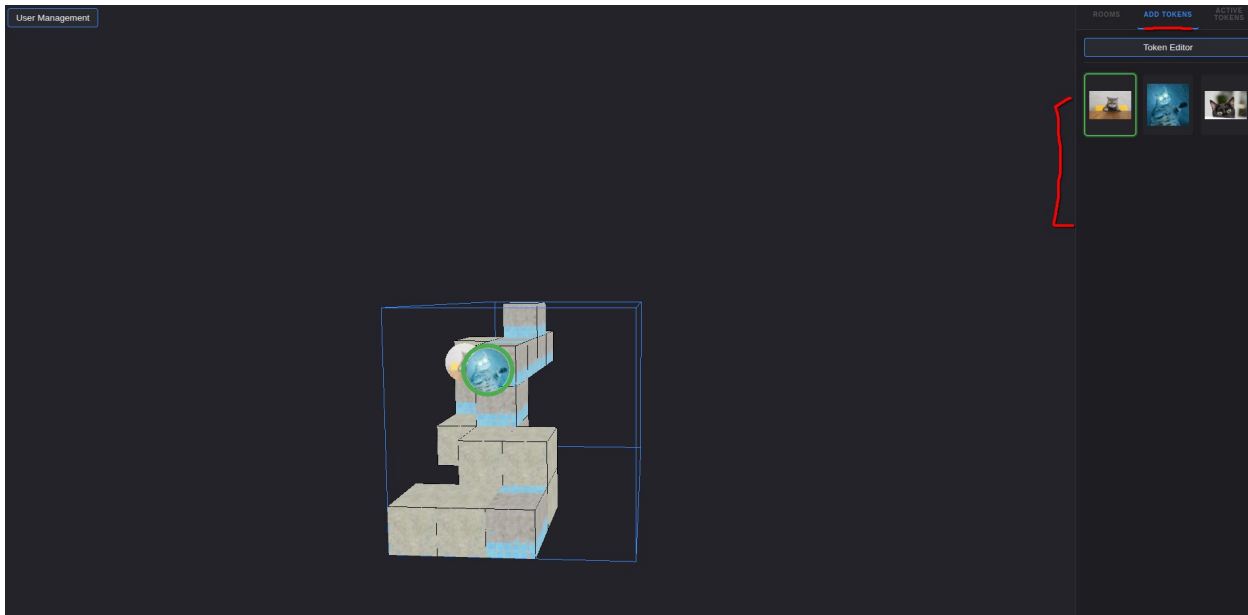
4.1: Selecting a room to be used

In order to change the selected room, the host needs to open the 'Rooms' tab in the right panel of the Home screen. The host just needs to click on the name of the room they have already created from the list of rooms (see Section 2). After clicking, the room will be loaded on every logged-in user's instance



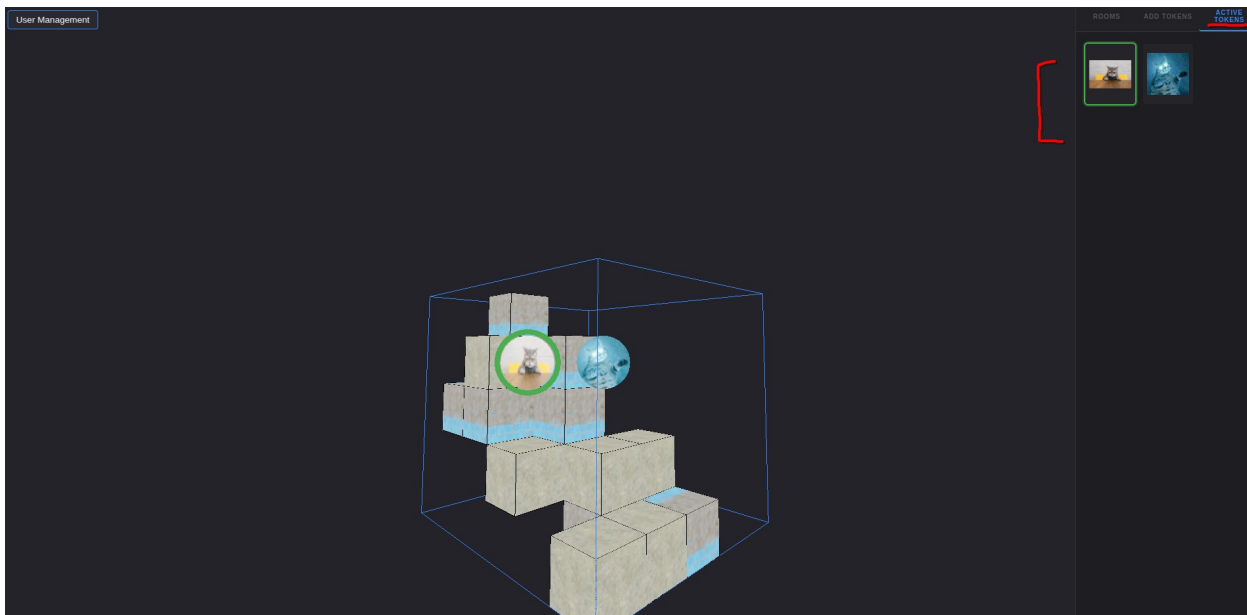
4.2 Adding tokens to the map

The Host needs to select the 'Add Tokens' tab from the sidebar on the right. Then, they need to click on the token they created (see Section 3). Then, they just need to click on the map where they want a new token instance to be placed.



4.3: Token Selection

In order to move a token around the map, any user needs to navigate to the 'Active Token' tab in the right pane. The user then needs to click the token they want to move. The token will get a green halo on the map.



4.4: Interacting with the map

The map has the following functions:

4.4.1: Rotating the camera

To rotate the camera, hold down the scroll wheel on the mouse. As the user moves their mouse around with this button pressed, the camera will rotate around the map. Upon release, the camera will maintain this angle.

4.4.2: Move the camera position

To move the camera's position, the user need to hold right click down and move the mouse. This will slide the camera around the 3D space. After releasing right click, the camera will remain in the position.

4.4.3: Moving a token

To change the position of a token, the user needs to left click and drag the token to the square they want. On release, the token will be moved to that position. It will move the selected token to the point of release. To select a token, see Section 4.3